Project 3 : Infrared Sensor Module

Introduction:

Pluto X2, Infrared Sensor module is an add-on, that is mostly applied to sense nearby object. In this project we are going to interface it with Pluto X2 and understand further scope

Challenge:

Build a drone that avoids collisions, due to surrounding objects.

Add-On used:

A IR sensor module consists of a IR led and IR light receiver which detects the reflection of IR rays from nearby objects. It returns value in digital mode that is 1’s and 0’s.

Connection:

Connect the IR_module on the RC port on Primus V5/X2 board

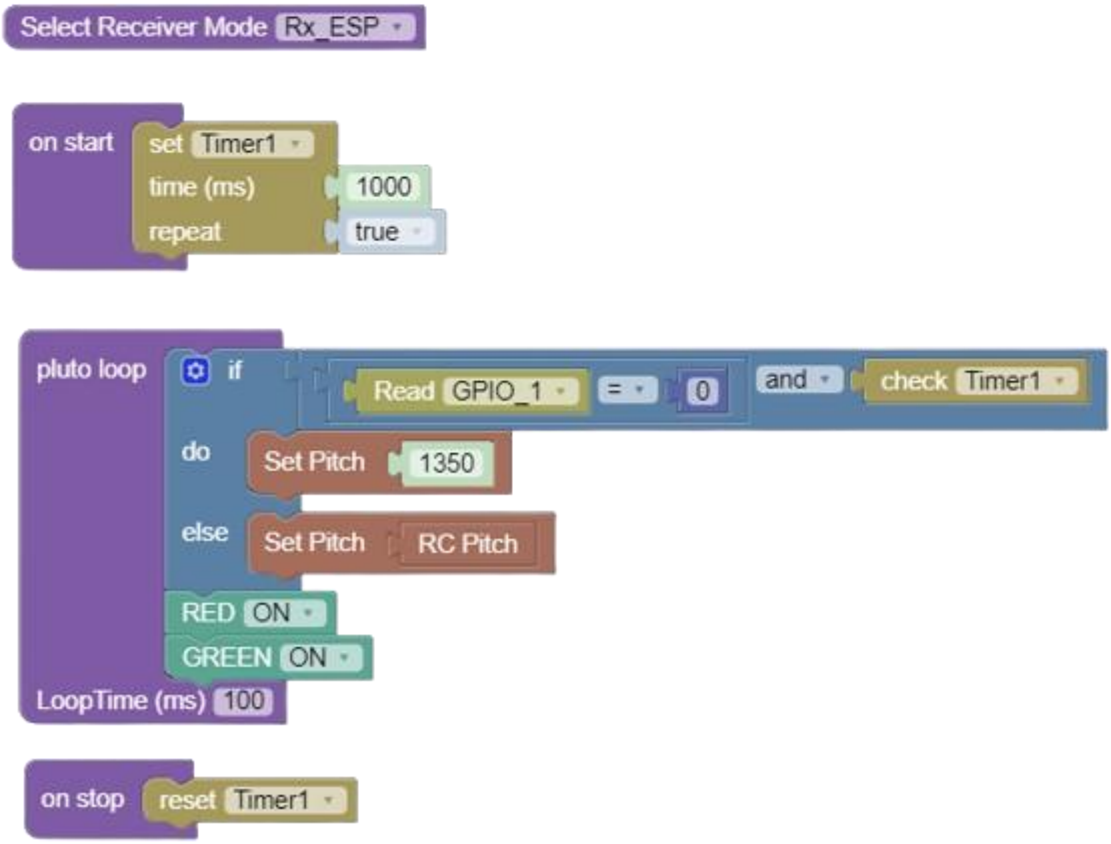
Primus V5/X2 pin	**Peripherals**
VCC on RC port	VCC on IR_module

GND on RC port	GND on IR_module
Signal on RC port	D0 on IR_module

Approaching the Problem:

We evaluate IR sensor readings , to know whether any drone has reached near any external object. If it is, we command it to move in opposite direction for certain amount of time.

Solution:



Explaination of Solution:

On Start Loop:
We set a repeatable timer of 1 second.

Pluto Loop:
In the loop, a constant yellow light glows, indicating developer mode is on. IR sensor under normal conditions outputs HIGH, but as it senses something (wall/external object) its output goes LOW. We check whether the IR has detected an object and timer1 is true. Timer1 is used here to make the drone follow "Set Pitch" command for 1 second. So that it is not in danger after timer goes off.

On stop Loop: Timer1 resets.

fig: IR Add-on with Pluto

Try further!

- Utilise multiple other sensors, like LDR to create a semi-autonomous drone.
- Add another IR sensor, for channel follower drone .
- Wall following drone.